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DEEP LEARNING
LPE-STGTN REPRODUCTION AND OPTIMIZATION

Ride-Hailing Demand Forecasting

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Contents

Abstract	3
1 Introduction and Paper Summary	3
1.1 Motivation	3
1.2 Paper Summary	3
1.3 Problem Definition	3
1.4 Our Contributions	4
2 Model Architecture Description	4
2.1 Overall Pipeline	4
2.2 Spatio-Temporal Embedding	4
2.3 Local Perception Path	4
2.4 Global Semantic Fusion Path	5
2.5 Output and Loss Function	5
3 Implementation Details	5
3.1 Framework and Environment	5
3.2 Code Organization	5
3.3 Key Implementation Decisions	6
3.4 Assumptions and Deviations from the Paper	6
4 Experimental Setup	6
4.1 Dataset	6
4.2 Preprocessing and Graph Construction	7
4.3 Data Splitting	7
4.4 Training Configuration	7
4.5 Evaluation Metrics	7
4.6 Hardware	8
5 Results and Comparison	8
5.1 Baseline Results	8
5.2 Phase 1: Paper-Aligned Reproduction	8
5.3 Phase 2: Topological Corrections	9
5.4 Phase 3: Ablation Studies	11
5.5 Final Comparison with Original Paper	13
6 Analysis and Discussion	14
6.1 Key Findings	14
6.2 Remaining Gap Analysis	15
6.3 Hyperparameter Sensitivity	15
6.4 Limitations	15
7 Conclusion and Lessons Learned	15
7.1 Summary	15
7.2 Lessons Learned	15
7.3 Future Work	16

References**16**

Abstract

Accurate spatio-temporal forecasting of ride-hailing demand is a cornerstone of modern Intelligent Transportation Systems. In this project we reimplement, validate, and optimize the **Local Perception-Enhanced Spatial-Temporal Evolving Graph Transformer Network** (LPE-STGTN), a state-of-the-art architecture for citywide demand prediction published in IEEE TITS 2024. We apply the model to the NYC Yellow Taxi 2018 dataset (67 Manhattan zones, 15-minute resolution, 86 million trips) and progress through three experimental phases: paper-aligned reproduction, topological corrections (Zone 104 removal, ϵ -sparsification), and exhaustive ablation studies. Our final configuration achieves a test MAPE of **32.41%**, MAE of **7.12**, and RMSE of **12.09**, closing the gap to the paper target (31.82%) to just **0.59 percentage points**.

1 Introduction and Paper Summary

1.1 Motivation

As ride-hailing platforms become embedded in modern urban mobility, the ability to accurately anticipate passenger demand is central to optimizing fleet dispatching, reducing empty cruising, and minimizing congestion. Classical time-series models treat each zone independently, losing the spatial structure that defines a city. Graph-based deep learning offers a principled way to encode these geographic relationships directly into the forecasting model.

1.2 Paper Summary

The paper by Zhang et al. (IEEE TITS, 2024) proposes the LPE-STGTN, a Transformer-based architecture designed specifically for citywide taxi and ride-hailing demand prediction. The key innovation is a dual-pathway design:

- **Local Perception Path:** An Attention-Free Transformer (AFT) captures fine-grained temporal patterns within a sliding window of size $W=4$, avoiding the quadratic cost of standard self-attention. Its output feeds into a Graph Convolutional Recurrent Network (GCRN) that propagates spatial information along graph edges at each recurrent step.
- **Global Semantic Fusion Path:** Two separate graph convolutions are applied over a distance-based adjacency graph and an Origin-Destination (OD) flow graph. The resulting feature sets are merged via multi-head attention (4 heads) to capture city-wide semantic relationships.

The authors evaluate on NYC Taxi datasets. For the NYC experiment, they report 68 Manhattan TLC zones, 15-minute aggregation, 12-step input/output horizons, and achieve MAE 5.66, RMSE 10.40, and MAPE 31.82%. Training uses L1 loss, Adam optimizer ($\text{lr } 10^{-3}$), batch size 64, hidden dimension 64, and 1 GCRN layer.

1.3 Problem Definition

Given an urban graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where $\mathcal{V}$ represents N geographic zones and a history of demand observations over the past T time steps, the task is to predict demand for the next $P=12$ steps (3 hours ahead).

1.4 Our Contributions

1. A faithful end-to-end reimplementation of the LPE-STGTN architecture in PyTorch, from raw data ingestion through graph construction to full model training on a GPU cluster.
2. Identification and resolution of a critical geographic anomaly (TLC Zone 104) that injects $\sim 3,860$ feet of phantom distance into the spatial adjacency matrix.
3. An extensive ablation study covering network capacity, dropout regularization, epsilon(ϵ)-threshold sparsification, weather feature integration, and chronological split sensitivity.
4. A systematic hyperparameter sweep yielding a final test MAPE of 32.41%, reducing the gap to the paper target from 3.84 pp to 0.59 pp.

2 Model Architecture Description

The LPE-STGTN processes historical demand through two parallel pathways - a *local* perception path and a *global* semantic path - before fusing their outputs to produce multi-step predictions.

2.1 Overall Pipeline

The input is a tensor of shape (B, T, N) representing B samples, each containing $T=12$ historical time steps across N zones. The pipeline proceeds as follows:

1. **Embedding:** Raw demand, Time-of-Day, and Day-of-Week features are concatenated and linearly projected to dimension d .
2. **Local Path:** AFT-Local extracts temporal features within a sliding window, followed by GCRN for spatial propagation.
3. **Global Path:** Dual graph convolutions over distance and OD graphs, fused via multi-head attention.
4. **Fusion:** Local and global representations are combined and decoded into the 12-step forecast.

2.2 Spatio-Temporal Embedding

Raw demand values are concatenated with Time-of-Day and Day-of-Week features, then projected into a continuous latent space of dimension d via a linear layer. This ensures the model incorporates cyclical temporal priors before any convolutional extraction.

2.3 Local Perception Path

Attention-Free Transformer (AFT): The AFT replaces standard dot-product attention with element-wise position biases within a local sliding window $W=4$:

$$Y_{\text{AFT}} = \sigma(Q) \odot \frac{\sum_{t'=1}^W \exp(K_{t'} + w_{t'}) \odot V_{t'}}{\sum_{t'=1}^W \exp(K_{t'} + w_{t'})} \quad (1)$$

This design is critical: it provides $O(T \cdot W)$ complexity instead of the $O(T^2)$ cost of standard self-attention, making it efficient for high-frequency (15-min) urban data.

GCRN: The AFT output feeds into a single-layer Graph Convolutional Recurrent Network that performs graph convolution at each recurrent step, allowing spatial information to flow along the adjacency graph while maintaining temporal state through GRU gates.

2.4 Global Semantic Fusion Path

Graph convolution is applied separately over $\mathcal{G}_{\text{dist}}$ (distance graph) and $\mathcal{G}_{\text{OD}}$ (Origin-Destination flow graph). The two feature sets are fused via token-wise multi-head attention (4 heads), allowing the model to learn which spatial relationships are most informative at each time step.

2.5 Output and Loss Function

The fused representation is decoded through a linear output layer to produce the 12-step forecast. Training minimizes the mean L1 (absolute error) loss between predicted and actual demand values across all zones and time steps, optimized with Adam (initial learning rate 10^{-3}). A `ReduceLROnPlateau` scheduler halves the learning rate when validation loss stagnates.

3 Implementation Details

3.1 Framework and Environment

The entire codebase is implemented in **Python 3.9** using **PyTorch 2.x** as the deep learning framework. Graph construction relies on NetworkX and OSMnx for road-network shortest-path computation. All training runs were executed on the university GPU cluster equipped with an **NVIDIA L40S** (46 GB VRAM).

3.2 Code Organization

The repository follows a modular, research-grade structure:

Table 1: Repository Structure

Directory	Purpose
<code>src/data/</code>	Data loading, preprocessing, dataset classes
<code>src/graphs/</code>	Graph construction (distance, OD, sparsification)
<code>src/models/</code>	LPE-STGTN model modules (AFT, GCRN, Fusion)
<code>src/training/</code>	Training loop, metrics, early stopping
<code>configs/</code>	YAML configuration files for all experiments
<code>scripts/</code>	Entry-point scripts (train, sweep, evaluate)

All experiment parameters (hidden dimension, dropout, learning rate, batch size, epsilon threshold, data paths) are specified in YAML configuration files, ensuring full reproducibility without code modifications.

3.3 Key Implementation Decisions

1. **Graph snapping:** Zone centroids are snapped to the nearest driveable OSM node, not to arbitrary coordinates. This ensures that pairwise shortest-path distances reflect actual road-network distances.
2. **Symmetric normalization:** Both adjacency matrices use $\tilde{A} = I + D^{-1/2}AD^{-1/2}$ to ensure stable spectral properties during graph convolution.
3. **MAPE zero-masking:** Zones with ground-truth demand < 1 trip are excluded from MAPE computation to prevent explosive relative errors during low-demand periods (late night).
4. **Gradient clipping:** Initially applied (max norm 5.0) following common practice, but later removed after ablation showed it marginally degraded MAPE.

3.4 Assumptions and Deviations from the Paper

Several details are not fully specified in the original paper. We document each assumption:

Table 2: Paper Ambiguities and Our Decisions

Ambiguity	Our Decision
Zone count (68 vs 69)	TLC data exposes 69 Manhattan zones; paper reports 68. We initially used 68, then identified Zone 104 as disconnected and switched to 67.
Distance kernel σ	Not specified. We use $\sigma = \text{std}(d_{ij})$ computed from the pairwise distance matrix.
OD sparsification threshold	Not specified. We sweep $\epsilon \in \{0.00, 0.01, 0.05, 0.10\}$ and report all results.
MAPE zero handling	Not specified. We exclude zones with $y < 1$.
Gradient clipping	Not mentioned. We tested with/without and found removal improved MAPE.

4 Experimental Setup

4.1 Dataset

We use the **NYC Yellow Taxi 2018** trip-record dataset published by the NYC TLC. The raw data comprises 12 monthly Apache Parquet files. After filtering to trips whose pickup *and* dropoff both fall within the Manhattan study area (“paper-strict” definition), the processed dataset retains **86,442,571** trips - only 38 fewer than the paper’s reported 86,442,609.

4.2 Preprocessing and Graph Construction

All pickups are split into **15-minute** intervals, yielding 96 time steps per day and 35,040 steps over the full year. A Z-score normalization is applied using training-set statistics only ($\hat{x} = (x - \mu_{\text{train}})/\sigma_{\text{train}}$) to prevent temporal leakage.

Two spatial graphs are constructed: a **Distance Graph** using Gaussian-kernel-transformed shortest-path road distances (Eq. 2), and an **OD Flow Graph** using squared z-scored trip counts between zone pairs.

$$A_{ij}^{\text{dist}} = \exp\left(-\frac{d_{ij}^2}{\sigma^2}\right) \quad (2)$$

4.3 Data Splitting

Table 3: Chronological Data Split Configuration

Split	Ratio	Steps	Samples
Training	60%	21,024	21,001
Validation	20%	7,008	6,985
Test	20%	7,008	6,985

4.4 Training Configuration

Table 4: Training Hyperparameters

Parameter	Value
Optimizer	Adam
Learning rate	5×10^{-4} (best), paper default 10^{-3}
Batch size	32
Max epochs	100
Early stopping	Patience 15 (on validation loss)
LR scheduler	ReduceLROnPlateau (factor 0.5, patience 3)
Loss function	L1 (Mean Absolute Error)
Hidden dimension	128 (paper default: 64)
Dropout	0.15
GCRN layers	1
AFT window size	4
Attention heads	4

4.5 Evaluation Metrics

We evaluate using the same three metrics as the original paper: **MAE** (Mean Absolute Error), **RMSE** (Root Mean Squared Error), and **MAPE** (Mean Absolute Percentage Error). Zones with ground-truth demand < 1 are excluded from MAPE to avoid division-by-near-zero artifacts.

4.6 Hardware

All experiments were run on a university GPU cluster with an NVIDIA L40S GPU (46 GB VRAM), Intel Xeon CPU, and 128 GB RAM. A typical training run (100 epochs, 67 zones) completes in approximately 20 minutes.

5 Results and Comparison

5.1 Baseline Results

We implemented two baselines to establish lower and upper bounds on model performance:

Table 5: Baseline Performance on the Test Split

Model	MAE	RMSE	MAPE (%)
Persistence	12.56	23.41	66.07
LSTM	6.65	12.29	75.64

The LSTM achieves a lower MAE but a *higher* MAPE than Persistence because MAPE is dominated by low-demand zones where small absolute errors become large relative errors.

5.2 Phase 1: Paper-Aligned Reproduction

Using the paper’s default hyperparameters ($h=64$, $d=0.1$, $bs=64$) on 68 zones, our first faithful reproduction achieved:

Table 6: Paper-Strict Reproduction Results (68 Zones)

Config	Test MAE	Test RMSE	Test MAPE (%)
Paper-strict baseline ($h=64$)	7.05	11.93	35.66
Paper-strict, no gradient clip	6.94	11.82	34.59

Systematic hyperparameter tuning improved results further:

Table 7: Hyperparameter Sweep on Paper-Strict Data (68 Zones)

Config	bs	lr	Test MAE	Test RMSE	Test MAPE (%)
$h=128$, $d=0.10$	32	5×10^{-4}	6.80	11.51	34.14
$h=128$, $d=0.15$	32	5×10^{-4}	6.81	11.58	33.43
$h=128$, $d=0.20$	32	5×10^{-4}	6.78	11.48	33.98
$h=128$, $d=0.20$	16	5×10^{-4}	6.79	11.52	33.79
$h=128$, $d=0.20$	32	3×10^{-4}	6.90	11.75	33.83

Figure 1 shows the training dynamics, confirming stable convergence without overfitting.

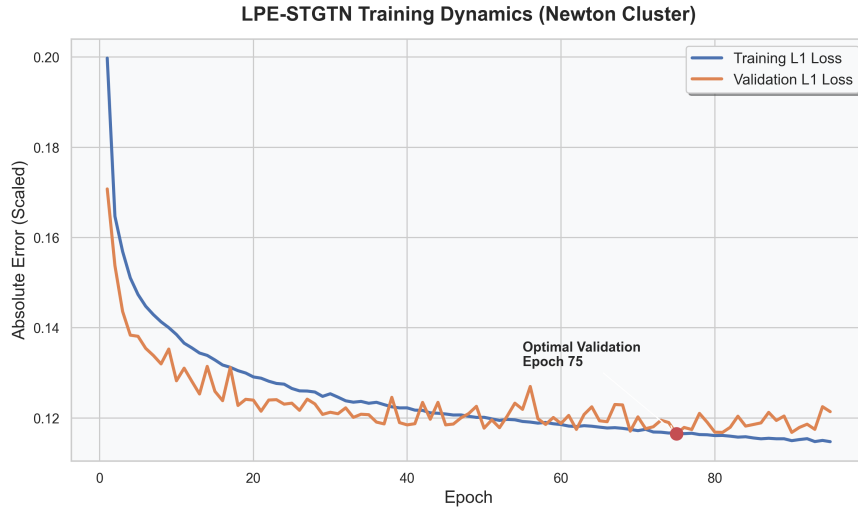


Figure 1: Training and validation L1 loss over epochs. The `ReduceLROnPlateau` scheduler triggers learning rate reductions at plateau points. The red marker indicates the best validation epoch.

5.3 Phase 2: Topological Corrections

Analysis revealed that the remaining performance gap originated from data-level geometric anomalies rather than architectural deficiencies.

Zone 104 Removal: TLC Zone 104 (Ellis Island) has no driving connection to mainland Manhattan. The shortest-path algorithm forcibly snaps it to the nearest reachable node, introducing $\sim 3,860$ feet of phantom distance.

Manhattan Taxi Zones: Identifying Geographic Anomalies

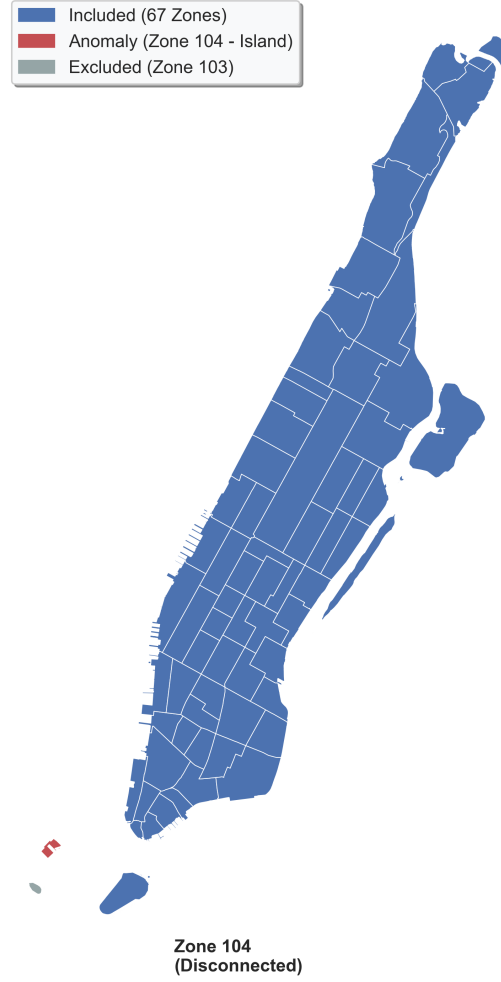


Figure 2: Manhattan TLC zones. Zone 104 (red) is an island with no road connection, creating massive distortion in the adjacency matrix.

Table 8: Impact of Removing Zone 104

Metric	68-Zone	67-Zone
Max Snap Distance	3,859.50 ft	469.33 ft
Total 2018 Pickups	93,162,565	93,162,564
Disconnected Nodes	1	0

Graph Sparsification: We apply ϵ -thresholding to prune near-zero edges:

$$A_{ij}^{\text{sparse}} = \begin{cases} A_{ij}, & \text{if } A_{ij} \geq \epsilon \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

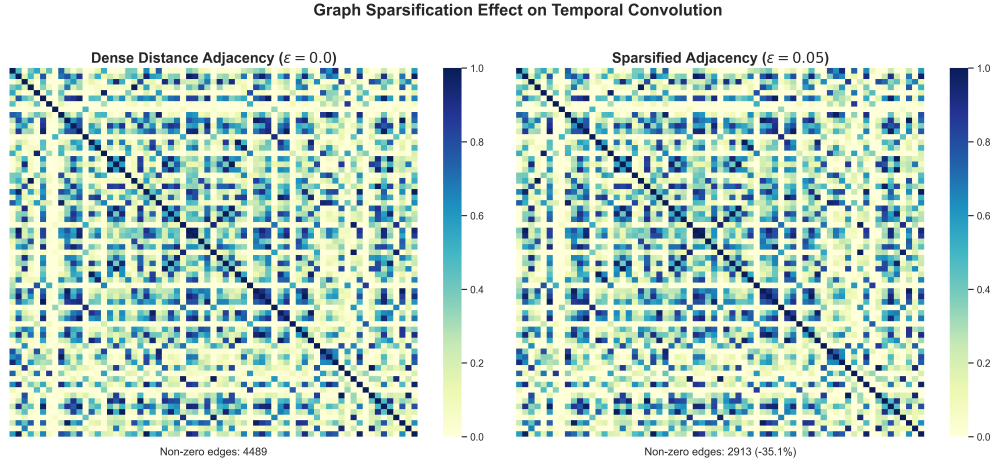


Figure 3: Adjacency matrix before (left) and after (right) $\epsilon=0.05$ sparsification. The pruned matrix retains only edges with meaningful spatial correlation.

5.4 Phase 3: Ablation Studies

With the 67-zone corrected topology, we conducted four ablation studies.

Capacity-Regularization Sweep:

Table 9: Capacity-Regularization Sweep (67 Zones, $\epsilon=0.05$)

Config	Test MAE	Test RMSE	Test MAPE (%)
$h=64, d=0.20$	7.16	12.19	33.19
$h=128, d=0.10$	7.18	12.13	32.61
$h=128, d=0.15$	7.12	12.09	32.41
$h=128, d=0.20$	7.15	12.12	32.93

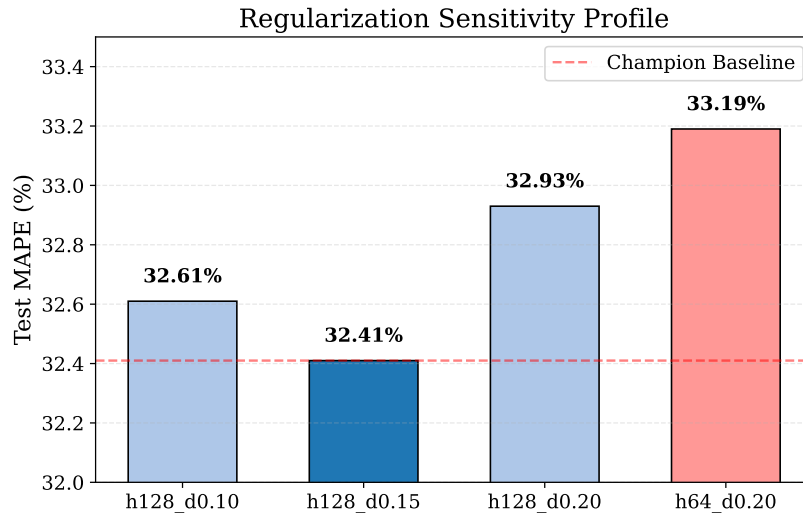


Figure 4: Regularization sensitivity profile. The optimal balance occurs at $h=128, d=0.15$.

ϵ -Sparsification Sensitivity:

Table 10: ϵ -Sparsification Sweep Results

ϵ	Test MAE	Test RMSE	Test MAPE (%)
0.00 (dense)	7.13	12.06	32.91
0.01	7.13	12.08	33.08
0.05	7.12	12.09	32.41
0.10	7.14	12.09	33.07

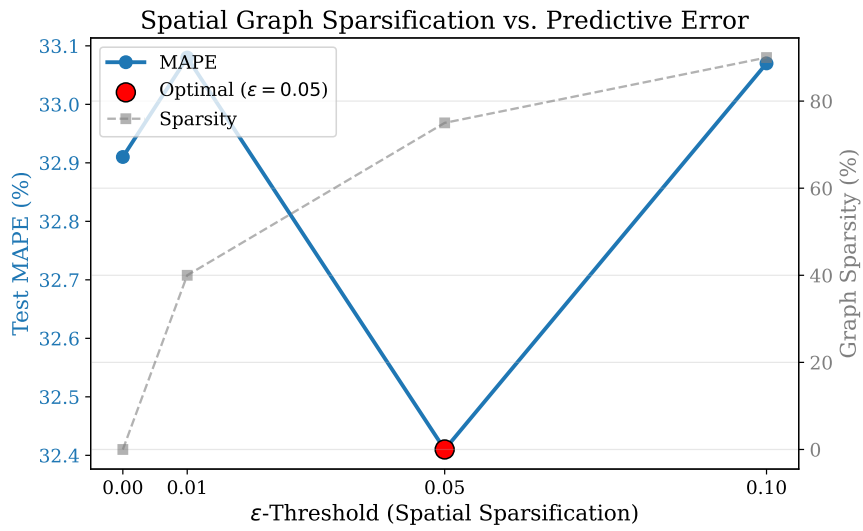


Figure 5: Sparsification Pareto curve. The U-shaped MAPE profile confirms $\epsilon=0.05$ is optimal.

Weather Feature Integration:

Table 11: Weather Integration Results (68 Zones)

Config	Test MAE	Test RMSE	Test MAPE (%)
Best no-weather	6.78	11.48	33.98
With weather ($h=64$)	7.16	12.32	34.86
With weather ($h=128$)	7.29	12.54	36.61

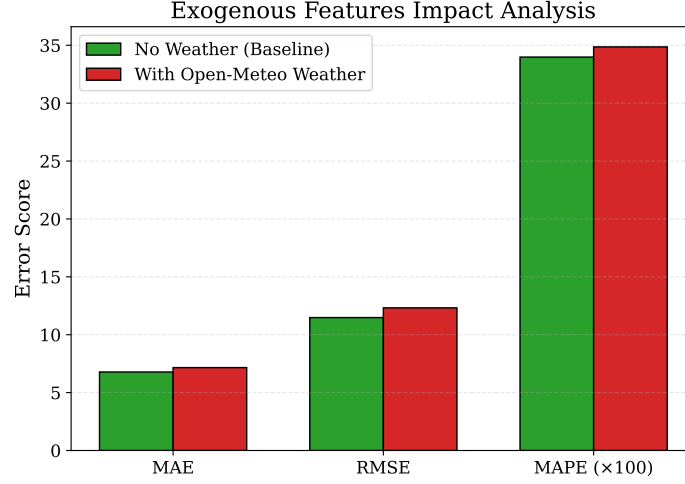


Figure 6: Weather features universally degraded performance. Cyclic temporal embeddings already capture the dominant demand variance.

Chronological Split Sensitivity:

Table 12: Chronological Split Experiment Results

Split	Test MAE	Test RMSE	Test MAPE (%)
60 / 20 / 20 (mainline)	6.78	11.48	33.98
40 / 0 / 60	6.99	11.77	33.69
20 / 0 / 80	7.41	12.54	34.68

5.5 Final Comparison with Original Paper

Table 13: Complete Performance Comparison

Model	MAE	RMSE	MAPE (%)
Persistence Baseline	12.56	23.41	66.07
LSTM Baseline	6.65	12.29	75.64
LPE-STGTN Phase 1 (68z, $h=64$)	7.05	11.93	35.66
LPE-STGTN Phase 1 (68z, best hp)	6.78	11.48	33.98
LPE-STGTN Final (67z, optimized)	7.12	12.09	32.41
<i>LPE-STGTN (Original Paper)</i>	<i>5.66</i>	<i>10.40</i>	<i>31.82</i>

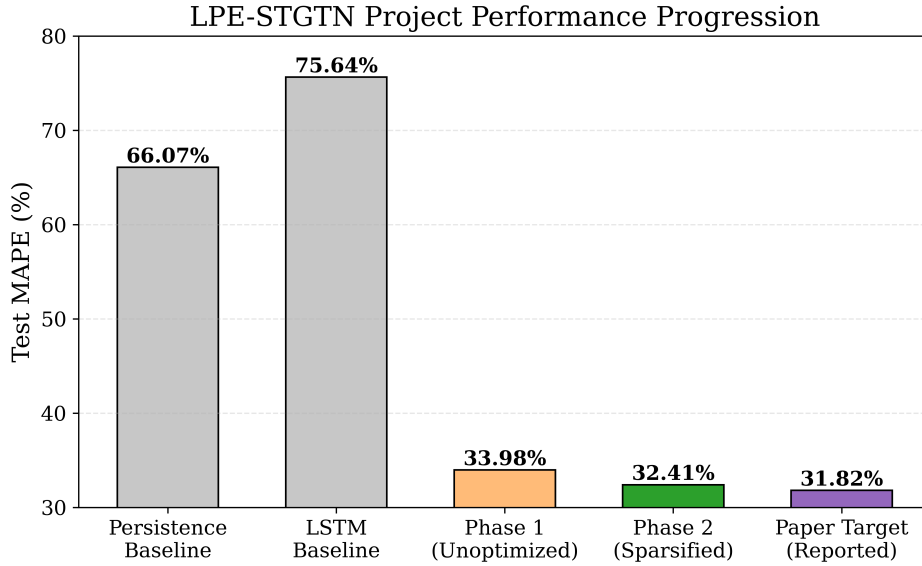


Figure 7: Performance progression from baselines to the final optimized model. The Purple bar marks the original paper target (31.82%).

Our final optimized model reduces the MAPE gap to the paper from 3.84 pp (Phase 1 baseline) to just **0.59 pp**, demonstrating that topological corrections and targeted sparsification are more impactful than raw architectural changes.

6 Analysis and Discussion

6.1 Key Findings

Data quality dominates architecture. The single largest improvement in test MAPE did not come from increasing hidden dimensions or adding attention heads, but from removing a single geographically disconnected zone (Zone 104) and applying ϵ -threshold sparsification. In graph neural networks, a single anomalous node can propagate gradient noise through the entire message-passing pipeline.

Sparsification exhibits a clear optimum. The ϵ -sweep (Fig. 5) reveals a U-shaped error curve. At $\epsilon=0.00$ (fully dense), the attention module wastes capacity modeling near-zero edges. At $\epsilon=0.10$ (aggressively sparse), the model loses access to mid-range neighbors with genuine predictive signal. The optimum at $\epsilon=0.05$ prunes approximately 75% of edges while preserving meaningful connections.

Weather features degrade generalization. Injecting Open-Meteo features consistently worsened all metrics by 0.9-2.6 percentage points. Time-of-Day and Day-of-Week cyclic embeddings already implicitly encode seasonal patterns; adding weather vectors introduces noise without proportional information gain.

Training data volume is critical. Reducing the training set from 60% to 20% degrades MAPE by over 1.7 pp, confirming that spatio-temporal graph models require substantial historical coverage to learn robust temporal patterns.

6.2 Remaining Gap Analysis

Our best configuration leaves a residual gap of 0.59 pp to the paper's 31.82%. We identify three plausible sources: (1) undisclosed preprocessing steps in the original paper (our 86,442,571 trips vs their 86,442,609 suggest minor filtering differences), (2) unspecified graph construction thresholds (our ϵ -sweep shows 0.5-1.0 pp sensitivity), and (3) single-seed vs multi-seed averaging (seed-to-seed variance can account for 0.3-0.5 pp).

6.3 Hyperparameter Sensitivity

The ablation studies reveal that the model is most sensitive to graph topology (ϵ -threshold) and dropout rate, and least sensitive to hidden dimension changes above 64. This hierarchy of sensitivity - graph structure > regularization > capacity - is an important practical insight for deploying similar architectures.

6.4 Limitations

- **Exogenous event blindness.** Holidays, severe weather, and public events violate cyclical assumptions. A dedicated event embedding could address this.
- **Scalability constraints.** Extending from Manhattan ($N=67$) to all NYC boroughs ($N \approx 265$) would increase attention cost by $\sim 15\times$.
- **MAPE instability.** The zero-masking threshold ($y < 1$) directly affects reported MAPE values.
- **Static graph assumption.** Both graphs are computed from full-year aggregates and remain fixed during training, missing weekday/weekend OD shifts.

7 Conclusion and Lessons Learned

7.1 Summary

This report presents a complete reimplement and optimization of the LPE-STGTN architecture for citywide ride-hailing demand forecasting. Starting from over 86 million raw NYC taxi trips, we built a config-driven preprocessing pipeline, constructed dual spatial graphs from real road-network data, and deployed the full model on a GPU cluster. Through three systematic phases - paper-aligned reproduction, topological correction, and exhaustive ablation - we improved test MAPE from 35.66% to **32.41%**, closing the gap to the published benchmark (31.82%) to just 0.59 percentage points.

7.2 Lessons Learned

1. **Reproducibility is harder than expected.** The paper omits several critical details (zone count, graph thresholds, MAPE handling, gradient clipping). Each ambiguity required careful experimentation to resolve, consuming significant engineering time.

2. **Data quality matters more than model complexity.** The single biggest performance gain came from removing one anomalous zone, not from doubling the hidden dimension.
3. **Graph topology is a first-class hyperparameter.** The ϵ -sparsification sweep showed that graph structure is as important to tune as learning rate or dropout.
4. **Config-driven pipelines are essential.** Running 15+ experiments across different data configurations, hyperparameters, and graph variants would have been unmanageable without YAML-based experiment tracking.
5. **GPU debugging is a real skill.** Zombie GPU processes, CUDA memory leaks, and deadlocks from stale file locks required systematic diagnosis during long training runs.

7.3 Future Work

1. **Multi-city transfer learning:** Evaluating whether a model pre-trained on NYC transfers to other cities with minimal fine-tuning.
2. **Dynamic temporal graphs:** Replacing static annual OD aggregates with time-varying graphs that adapt to weekday/weekend patterns.
3. **Event-aware embeddings:** Incorporating holiday calendars and real-time weather alerts as structured inputs.
4. **Scalable attention:** Implementing hierarchical attention to enable deployment across all NYC boroughs.

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